

COS40007 Design project 2026 – DRAFT – Due Week 13

Fricaday, 11:59PM, Interview Week 14

COS40007 students are expected to undertake a design project on a focused topic of AI for Engineering. Students are encouraged to collect data by their own if available or create their own. This document contains a summary description of each project.

A. Grouping Rules

- A group needs to be formed to complete the design project
- The group should contain 4-5 students.
- Group are expected to produce the following outcome
 - data labelling
 - data exploration and data pre-processing
 - training and validation of machine learning/deep learning models
 - evaluation of machine learning/deep learning models
 - an AI demonstrator of final selected model
 - project presentation
 - a final report

B. Rubric, Report and Project Progress

- A rubric for the design project and outline of final report is available on CANVAS
- Each team is required to meet with Dr Joel to check on progress at least once. A form is available on CANVAS to document this meeting.

C. Project Topics

- The topics capture the overall project goals to solve AI problem in Engineering, but do not define specifications like portfolio tasks on how the project should be undertaken.
- You should apply your knowledge and skills you acquired in this unit to complete your project
- Project groups have their own freedom to choose technology component

D. Presubmission Consultation

- It is **mandatory** to consult with your tutor at least once before Week 11 on your project selection, project progression and breakdown of task for each member.
- A form will be available on CANVAS for you to use to record this meeting that must be signed by your tutor
- This form if failed to be submitted with your Design Project will cause a 20% penalty of your group

E. Marking Rubric

Marking criteria distribution will be adapted appropriately according to project chosen

Task	Marks	Level 5 (Advanced)	Level 4 (Proficient)	Level 3 (Developing)	Level 2 (Emerging)	Level 1 (Beginning)
1. Data Collection, Labelling & Preprocessing (ULO1)	10	9–10: Large, diverse self-collected dataset; accurate labels; clear class definitions; strong preprocessing & augmentation with justification	7.0–8.9: Good dataset; mostly correct labels; some preprocessing	4.5–6.9: Moderate dataset; some label issues; limited preprocessing	3.0–4.4: Small dataset; inconsistent labels; minimal preprocessing	0–4.9: Very small/irrelevant dataset; poor/no labelling
2. Training & Validation (ULO2)	5	4–5.0: Multiple models well-trained with clear understanding of algorithm (e.g., YOLO); hyperparameters justified; proper validation strategy	3–3.9: Correct training with some tuning; reasonable understanding	2.0–2.9: Basic training; limited explanation of model	1.5–1.9: Weak setup; unclear understanding	0–1.49: Model not properly trained
3. Detection on Unseen Data (Generalisation) (ULO3)	5	4.25–5.0: Strong generalisation; handles variation; clear original justification of performance and limitations	3.6–4.24: Works reasonably well; some justification	3.25–3.5: Limited generalisation; weak explanation	2.1–3.24: Poor performance; minimal justification	0–2: Fails on unseen data
4. Evaluation Metrics & Discussion(ULO3)	10	8.5–10: Uses multiple metrics (eg: mAP, IoU, precision/recall); deep analysis; strong justification of chosen methods and comparisons	7–8.4: Appropriate metrics; good discussion	6.1–6.9: Some metrics; limited analysis	5.0–6.4: Basic metrics only; weak explanation	0–4.9: No proper evaluation
5. User Interface / System Implementation (ULO1)	5	4.25–5.0: Functional, intuitive system; well integrated; demonstrates engineering design	3.75–4.24: Working system with minor issues	3.25–3.74: Basic system; limited usability	2.5–3.24: Minimal implementation	0–2.49: No working system
6. Interview (Individual Understanding) (ULO1-3)	15	12.75–15: Deep understanding across dataset, model, and evaluation; strong justification; explains trade-offs & limitations clearly	11.25–12.74: Good understanding with minor gaps	9.75–11.24: Basic understanding; limited depth	7.5–9.74: Weak understanding; struggles to explain	0–7.49: Cannot explain own work

Theme 1: Smart Campus / Campus Monitoring System

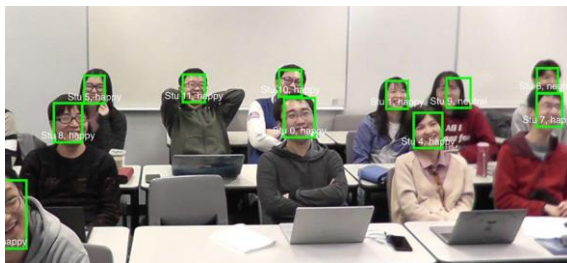
AI areas: Deep Learning, Machine Learning, Object classification, Transfer Learning

Format of Data: Classroom images, webcam images of faces

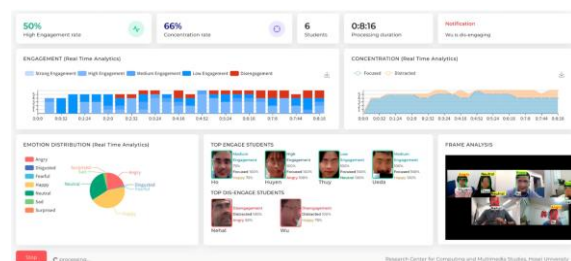
Topic: AI Classroom Engagement Tracker using Object Detection

Description: This project focuses on designing and developing an AI-based system that analyzes classroom environments to detect and quantify student engagement levels. Students are required to build an object detection model capable of identifying individuals within a classroom and classifying their engagement states (e.g., attentive, distracted, or drowsy) based on visual cues. The system should process image or video input and produce meaningful outputs such as annotated frames and engagement summaries. Students are encouraged to consider real-world constraints such as varying lighting conditions, camera angles, and occlusions. The project may also be extended to include real-time processing or a simple dashboard for visualization.

This task emphasizes not only model performance but also dataset design, annotation quality, and justification of chosen AI techniques, aligning with practical engineering applications in smart education systems.



(a) Example of detection



(b) Example of possible dashboards

Figure 1: Example of Class Monitoring Solution

Key Question to Answer

1. What challenges did you face during annotation, and how did you ensure **label consistency and quality**?
2. What challenges did you encounter during training (e.g., overfitting, class imbalance), and how did you address them?
3. How would your system scale to different classroom settings (e.g., lecture halls vs small rooms)?

Data source/format

You are tasked to source/create your own dataset. A minimum of three classes is required (min 200 images/annotations per class). Consider different angles, colours, lighting of those particular classes as well.

Input Data:

- Images or video frames (RGB)
- Classroom scenes (single or multiple students)

Annotations:

- Bounding boxes around each student
- Class labels such as:
 - *Attentive*
 - *Distracted*
 - *Drowsy*

Annotation Format:

- YOLO format (.txt) or COCO JSON

Data Source Options:

- Self-collected dataset (webcam / recorded lectures/your own)
- Public datasets (if applicable, but must be adapted)

Output Data:

- Detected bounding boxes + predicted engagement labels
- Optional: engagement statistics (e.g., % attentive students)

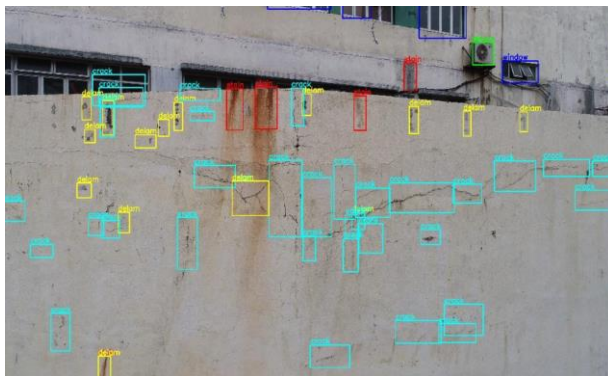
Theme 2: Smart Campus / Building Facilities Management

AI areas: Deep Learning, Machine Learning, Object classification, Counting

Topic: AI-Based Structural Defect Detection for Infrastructure Monitoring

Description: This project focuses on developing an AI-based system to detect structural defects in infrastructure such as cracks in concrete, road damage, or surface deterioration. Students will design and train an object detection model to identify and localize defects from images or video input. The system should be capable of assisting engineers in early detection of structural issues, improving maintenance efficiency and safety. Students are expected to consider real-world challenges such as varying lighting conditions, noise, surface textures, and different scales of defects.

Advanced implementations may include segmentation for precise crack detection or severity estimation based on defect size.



(a) Example of Crack Prediction



(b) Example of Facilities Management Dashboard

Figure 2: Example of Facilities Monitoring Solution

Key Question to Answer

1. How was your dataset collected, and how did you ensure variation in surface types, lighting, scale, and viewpoints?
2. How did you configure training (e.g., epochs, learning rate, image size), and what impact did these parameters have?
3. How reliable is your model for engineering decision-making, and what are its limitations?

Data source

You are tasked to source/create your own dataset. A minimum of three classes is required (min 200 images/annotations per class). Consider different angles, colours, lighting of those particular classes as well. Consider different angles, colours, lighting, indoor/outdoor parking. Public datasets (crack detection datasets), Self-collected (campus roads, walls, structures)

Input Data:

- Images or videos of infrastructure (bridges, roads, buildings)

Annotations:

- Bounding boxes or masks for (examples):
 - Cracks
 - Spalling
 - Corrosion
 - Potholes
 - Paint degradation

Annotation Format:

- YOLO (.txt) or COCO JSON

Data Sources:

- Public datasets (crack detection datasets)
- Self-collected (campus roads, walls, structures)

Output:

- Detected defects + severity indication (optional)

Theme 3: Smart Government

AI areas: Machine Learning, Predictive analytics, Time Series Forecasting or Clustering

Format of Data: Raw data from Malaysia's government repository or Other Countries.

Topic: Analyzing Malaysian Public Sector Data Using Time Series or Clustering Techniques

Description: Open challenge and requires meeting and approval from Dr Joel. Students are tasked with selecting a dataset from the OpenDOSM Data Catalogue that is suitable for either time series analysis or clustering. The objective is to derive meaningful insights that can inform policy decisions, economic strategies, or social programs in Malaysia.

The chosen dataset should be rich in temporal or categorical data to facilitate the application of AI and statistical techniques. Students will preprocess the data, apply appropriate analytical methods, and interpret the results in the context of Malaysian socio-economic development.

Key Question to Answer

1. Which dataset from OpenDOSM was selected, and what are its key features?
2. What time series or clustering methods were applied, and why were they chosen?
3. What patterns, trends, or groupings were discovered through the analysis?
4. What patterns, trends, or groupings were discovered through the analysis?

Input and Output of final AI model

- A cleaned and preprocessed dataset from OpenDOSM.

Output:

1. **Analytical Results:** Visualizations (e.g., line graphs, cluster plots) illustrating the findings.
2. **Interpretation:** A comprehensive report detailing the **methodology**, results, and potential implications.

Data Source

Students are tasked with selecting a dataset from the [OpenDOSM Data Catalogue](#) that is suitable for either time series analysis or clustering. The objective is to derive meaningful insights that can inform policy decisions, economic strategies, or social programs in Malaysia. Approval from Dr Joel must be sought before finalizing your choice of data. Alternatively you can select and compare other countries data as well.

Theme 4: Open Topic

AI areas: Open to suggestions must be related to areas covered in COS40007

Format of Data: Dependent on topic proposed.

Topic: Open to suggestions

Description: Open to suggestions. Your team needs to discuss and obtain approval with Dr Joel before beginning this project.

Key Question to Answer

- Open to suggestions

Input and Output of final AI model Input:

- Open to suggestions

Output:

- Open to suggestions

Data Source

- Open to suggestions